

Kindergarten Management - Data warehouse design

Business process

The Data Warehouse is designed for the **meal consumption business process** in kindergarten management. This process tracks meal preparation, distribution and waste at a group level, categorized by children's age.

Relational Database Schema

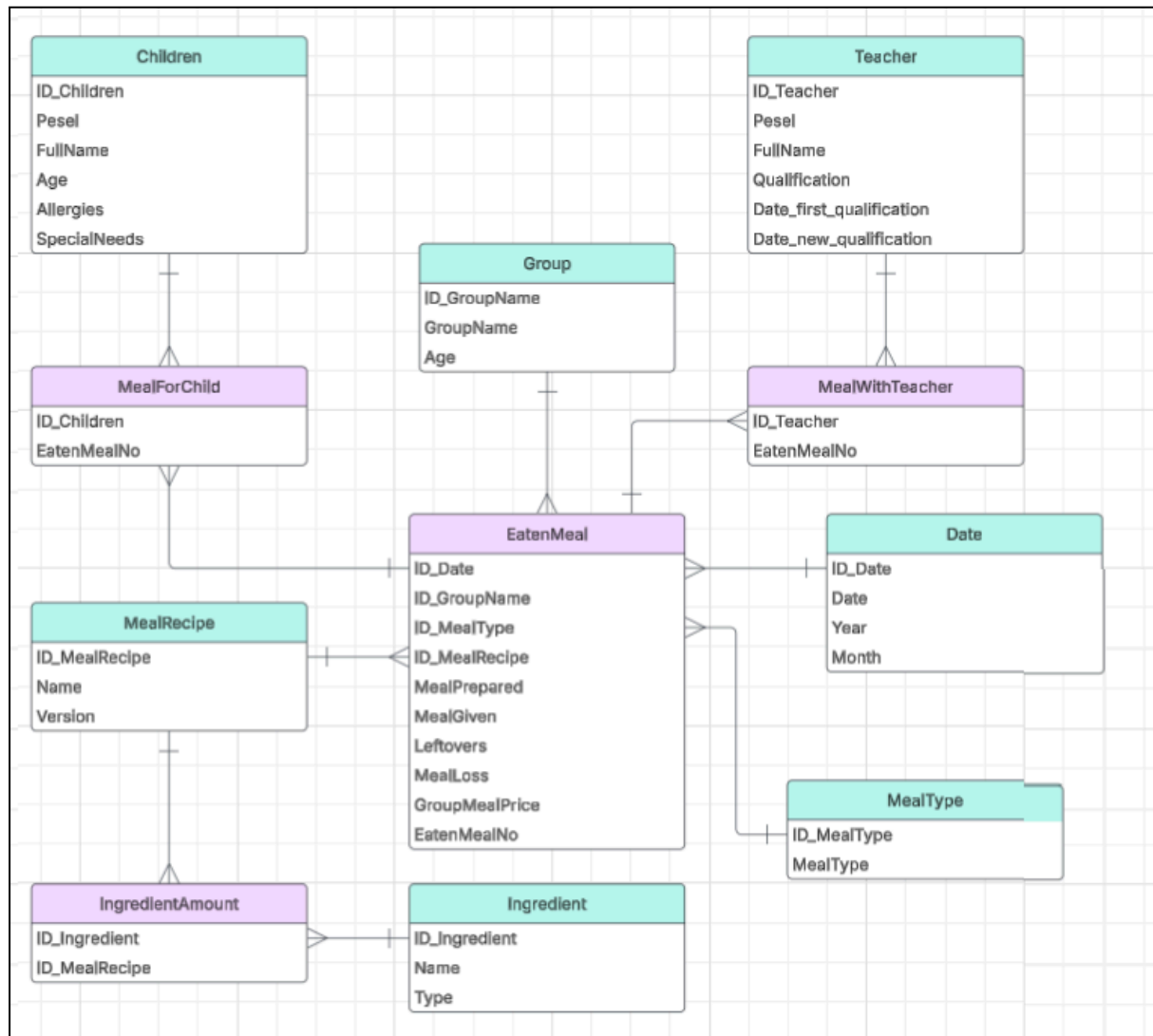


TABLE NAME	ATTRIBUTE	ATTRIBUTE TYPE	DESCRIPTION
EatenMeal (FACT TABLE)	One tuple describes one fact of an eaten meal.		
	ID_Date	Numeric	FK Date

	Meal date.		
	ID_GroupName	Numeric	FK GroupName Specific group.
	ID_MealType	Numeric	FK MealType Type of meal.
	ID_MealRecipe	Numeric	FK MealRecipe Recipe.
	MealPrepared	Numeric	Amount of meals prepared
	MealGiven	Numeric	Amount of meals given to children
	Leftovers	Numeric	Amount of meals not eaten
	MealLoss	Money	Loss equal to amount of ingredient * leftover quantity * price per unit - for every ingredient in this meal
	GroupMealPrice	Money	total cost of ingredients for the prepared meals
	EatenMealNo	Numeric	surrogate key, generated by DW
IngredientAmount (FACT TABLE)	One tuple describes one fact of usage of a specific ingredient in a specific recipe.		
	ID_Ingredient	Numeric	FK Ingredient Ingredient used.
	ID_MealRecipe	Numeric	FK MealRecipe In which recipe.
Ingredient (DIMENSION TABLE)	One tuple describes one ingredient, of a specific type.		
	ID_Ingredient	Numeric	PK
	Name	varchar (20)	Name of ingredient, BK
	Type	varchar (20)	Type of ingredient, allowed values: Dairy,

	Meat, Fruit, Vegetable, Side/other		
MealRecipe (DIMENSION TABLE)	One tuple describes one, specific version of the recipe.		
	ID_MealRecipe	Numeric	PK
	Name	varchar (30)	Name of recipe, BK
	Version	varchar (10)	Version of recipe ('small', 'big' ...), BK
Group (DIMENSION TABLE)	One tuple describes one group, in a specific age category.		
	ID_GroupName	Numeric	PK
	GroupName	varchar (20)	Name of group, BK
	Age	varchar (10)	Age category, allowed values: 3-4yr, 4-5yr, 5-6yr
MealType (DIMENSION TABLE)	One tuple describes one meal type.		
	ID_MealType	Numeric	PK
	MealType	varchar (20)	BK, Type of meal, allowed values: breakfast, lunch, snack, dinner
Date (DIMENSION TABLE)	One tuple describes one day.		
	ID_Date	Numeric	PK
	Date	Date	Date
	Year	4 digits	Year

	Month	varchar(10)	Month. Allowed values: January, February, March, April, May, June, July, August, September, October, November and December.
MealWithTeacher (FACT TABLE)	One tuple describes one meal with one specific teacher.		
	ID_Teacher	Numeric	FK, Teacher.
	EatenMealNo	Numeric	FK, Eaten Meal.
Teacher (DIMENSION TABLE SCD 2)	One tuple describes one teacher.		
	ID_Teacher	Numeric	PK
	Pesel	11 digits	BK, pesel of teacher
	FullName	varchar(60)	Name and surname of the teacher
	Qualification	varchar(100)	Qualification of the teacher, education status
	Date_first_qualification	date	Date of first qualification
	Date_new_qualification	date	If qualification changes, fill in this date
MealForChild (FACT TABLE)	One tuple describes one meal for a specific child.		
	ID_Children	Numeric	FK, Children
	EatenMealNo	Numeric	FK, Eaten Meal
Children (DIMENSION TABLE)	One tuple describes one child.		
	ID_Children	Numeric	PK
	Pesel	11 digits	BK, pesel of a child
	FullName	varchar(60)	Name and surname of the child
	Age	varchar (10)	Age category, allowed values: 3yr, 4yr, 5yr, 6yr

	Allergies	varchar(3)	Does child have any allergies, possible answers: 'Yes', 'No'
	SpecialNeeds	varchar(3)	Does child have any special needs, possible answers: 'Yes', 'No'

Dimensional model

Fact definitions

Fact 1 - Eaten meal: Consumption of a meal (specific meal recipe), by a specific group (we recognize group by age of children enrolled into them) on a given date and of a specific type (like on breakfast or diner).

Fact 1 table: EatenMeal

Granularity:

- a specific date
- a specific meal type
- a specific meal recipe
- a specific group

Measures and aggregation functions:

- number of meals prepared - SUM (MealPrepared)
- number of meals given - SUM (MealGiven)
- number of leftovers - SUM (Leftovers)
- meal loss - SUM (MealLoss)
- meal price for the whole group - SUM (GroupMealPrice)
- WastePercentage = number of leftovers/ number of meals prepared (calculated measure)

Fact 2 - Amount of ingredient: Usage of ingredient in a specific meal

Fact resulting from the "many to many" relationship.

Fact 2 table: IngredientAmount

Granularity:

- a specific ingredient
- a specific meal recipe

Fact 3 - Meal for child: A child that eats a specific meal

Fact resulting from the "many to many" relationship.

Fact 3 table: MealForChild

Granularity:

- a specific child
- a specific meal eaten

Fact 4 - Meal with teacher: A teacher that supervises eating a specific meal

Fact resulting from the "many to many" relationship.

Fact 4 table: MealWithTeacher

Granularity:

- a specific teacher
- a specific meal eaten

Assumptions:

- When calculating meal losses, we do not take into account meals that have been discarded due to the child's absence (Meal Prepared - Meal Given)
- We don't recognize specific children, we treat them as a group (depending on age)
- Each recipe can be prepared from more than 2 ingredients (not stated exactly from how many)
- Recipe differs only based on amount of ingredients (the same ingredients, only different amount - version of recipe)
- MealsPrepared is never equal to 0 (WastePercentage is always a valid expression)

Dimension definitions

Dimensions for Fact 1 - Eaten meal:

DIMENSION/DIMENSION ATTRIBUTE	TABLE/COLUMN	TYPE
GROUP	Group	Dimension
NAME OF GROUP	Group.GroupName	Dimension attribute
AGE	Group.Age	Dimension attribute
DATE HIERARCHY	• Date.Year •• Date.Month ••• Date.Date	Hierarchical dimension
DATE	Date	Dimension
YEAR	Date.Year	Dimension attribute
MONTH	Date.Month	Dimension attribute
MEAL RECIPE	MealRecipe	Dimension

NAME OF RECIPE	MealRecipe.Name	Dimension attribute
VERSION OF RECIPE	MealRecipe.Version	Dimension attribute
MEAL TYPE	MealType	Dimension
INGREDIENT	Ingredient	Dimension
NAME OF INGREDIENT	Ingredient.Name	Dimension attribute
TYPE OF INGREDIENT	Ingredient.Type	Dimension attribute
CHILDREN	Children	Dimension
CHILD PESEL	Children.Pesel	Dimension attribute
CHILD FULL NAME	Children.FullName	Dimension attribute
AGE CATEGORY	Children.Age	Dimension attribute
ALLERGIES	Children.Allergies	Dimension attribute
SPECIAL NEEDS	Children.SpecialNeeds	Dimension attribute
TEACHER	Teacher	Dimension
TEACHER PESEL	Teacher.Pesel	Dimension attribute
TEACHER FULL NAME	Teacher.FullName	Dimension attribute
QUALIFICATION	Teacher.Qualification	Dimension attribute
DATE 1 QUALIFICATION	Teacher.Date_first_qualification	Dimension attribute
DATE 2 QUALIFICATION	Teacher.Date_new_qualification	Dimension attribute

Dimensions for Fact 2 - Amount of ingredient:

DIMENSION/DIMENSION ATTRIBUTE	TABLE/COLUMN	TYPE
MEAL RECIPE	MealRecipe	Dimension
NAME OF RECIPE	MealRecipe.Name	Dimension attribute
INGREDIENT	Ingredient	Dimension
NAME OF INGREDIENT	Ingredient.Name	Dimension attribute
TYPE OF INGREDIENT	Ingredient.Type	Dimension attribute

Dimensions for Fact 3 - Meal for child:

DIMENSION/DIMENSION ATTRIBUTE	TABLE/COLUMN	TYPE
CHILDREN	Children	Dimension
CHILD PESEL	Children.Pesel	Dimension attribute
CHILD FULL NAME	Children.FullName	Dimension attribute
AGE CATEGORY	Children.Age	Dimension attribute
ALLERGIES	Children.Allergies	Dimension attribute
SPECIAL NEEDS	Children.SpecialNeeds	Dimension attribute

Dimensions for Fact 4 - Meal with teacher:

DIMENSION/DIMENSION ATTRIBUTE	TABLE/COLUMN	TYPE
TEACHER	Teacher	Dimension
TEACHER PESEL	Teacher.Pesel	Dimension attribute
TEACHER FULL NAME	Teacher.FullName	Dimension attribute
QUALIFICATION	Teacher.Qualification	Dimension attribute
DATE 1 QUALIFICATION	Teacher.Date_first_qualification	Dimension attribute
DATE 2 QUALIFICATION	Teacher.Date_new_qualification	Dimension attribute

Checking the feasibility of queries based on the multidimensional model

Why was there an increase or decrease in meal waste across 3 months?

1. Compare total meals prepared vs. meals given vs. leftovers for each meal type in the past three months.

Measure: Number of meals prepared, Number of meals given, Number of leftovers

Dimension: Meal Type (dimension attributes: Meal Type)

Dimension: Date (dimension attributes: Month, Year)

2. How does the total cost of ingredients in a single portion affect the meal waste?

Measure: Meal price for the whole group, Number of leftovers

Dimension: Meal Recipe (dimension attributes: Name of recipe)

Dimension: Date (dimension attributes: Month, Year)

3. Compare waste levels for meals with different portion sizes across 3 months.

Measure: WastePercentage (number of leftovers/ number of meals prepared)

Dimension: Meal Recipe (dimension attributes: Name of recipe, Version)

Dimension: Date (dimension attributes: Month, Year)

4. Identify meal types with the highest increase/decrease in waste over the last three months.

Measure: WastePercentage (number of leftovers/ number of meals prepared)

Dimension: Meal Type (dimension attributes: Meal Type)

Dimension: Date (dimension attributes: Month, Year)

5. Compare total cost of ingredients vs. the number of meals prepared.

Measure: Meal price for the whole group, Number of meals prepared

Dimension: Meal Recipe (dimension attributes: Name of recipe)

Dimension: Date (dimension attributes: Month, Year)

What is the effect of meal portion and ingredient adjustments on food waste reduction?

1. Does having more/fewer products of a certain type affect the waste reduction?

Measure: WastePercentage (number of leftovers/ number of meals prepared)

Dimension: Ingredient (dimension attributes: Type of ingredient)

Dimension: Date (dimension attributes: Month, Year)

2. What is the most common ingredient that can be found in a meal with the highest waste level?

Measure: WastePercentage (number of leftovers/ number of meals prepared)

Dimension: Ingredient (dimension attributes: Name of ingredient, Type of ingredient)

Dimension: Meal Recipe (dimension attributes: Name of recipe)

3. Calculate the cost of meals wasted due to leftovers from the last 3 months and calculate rate of change.

Measure: Meal loss

Dimension: Date (dimension attributes: Month, Year)

Dimension: Meal Type (dimension attributes: Meal Type)

4. Identify if removing or substituting ingredients affected waste levels.

Measure: WastePercentage (number of leftovers/ number of meals prepared)

Dimension: Ingredient (dimension attributes: Name of ingredient, Type of ingredient)

Dimension: Meal Recipe (dimension attributes: Name of recipe)

Dimension: Date (dimension attributes: Month, Year)

5. Compare waste levels for meal types that were adjusted vs. those that were not.

Measure: WastePercentage (number of leftovers/ number of meals prepared)

Dimension: Meal Type (dimension attributes: Meal Type)

Dimension: Meal Recipe (dimension attributes: Name of recipe, Version)

Checking if there are Data in the Data sources needed to fill the Data warehouse

Table name	Column	Source
EatenMeal	One tuple describes one fact of an eaten meal.	
	ID_Date	Date Id. Foreign key from Date dimension table. Based on Date stored in MealSchedule table in Kindergarten Management System source.
	ID_GroupName	Group name Id. Foreign key from Group dimension table. Based on GroupName stored in Group table in Kindergarten Management System source.
	ID_MealType	Meal type Id. Foreign key from MealType dimension table. Based on MealType stored in MealSchedule table in the Kindergarten Management System source.
	ID_MealRecipe	Meal recipe Id. Foreign key from MealRecipe dimension

	e	table. Taken from ID stored in MealRecipe table in Kindergarten Management System source.
	MealPrepared	Number of meals prepared taken from MealsPrepared column from MealConsumption table in Kindergarten Management System source.
	MealGiven	Number of meals given taken from MealsGiven column from MealConsumption table in Kindergarten Management System source.
	Leftovers	Number of leftovers taken from Leftovers column from MealConsumption table in Kindergarten Management System source.
	MealLoss	Cost of each ingredient of every wasted meal. Calculated as $\text{SUM}(\text{PricePerUnit} * \text{IngredientAmount} * \text{Amount} * \text{MealConsumption} * \text{Leftovers})$. PricePerUnit - taken from Column 5 from Order list CSV file source. The rest is taken from Kindergarten Management System, structured as <i>table.attribute</i> .
	GroupMealPrice	Total cost of ingredients for the prepared meals, calculated as $\text{SUM}(\text{PricePerUnit} * \text{IngredientAmount} * \text{Amount} * \text{MealConsumption} * \text{MealsPrepared})$. PricePerUnit - taken from Column 5 from Order list CSV file source. The rest is taken from Kindergarten Management System, structured as <i>table.attribute</i> .
	EatenMealNo	Number of eaten meal. Surrogate key - generated by database.
Date	One tuple describes one day. All the data in this table are generated tuple by tuple based on any calendar, before ETL process.	
Group	One tuple describes one group, in a specific age category.	
	ID_GroupName	Group name Id. Surrogate key - generated by the database.
	GroupName	Kindergarten group name. Taken from GroupName from Group table in Kindergarten Management System source.
	Age	Age category. Allowed values: 3-4yr, 4-5yr, 5-6yr. Determined from GroupName from Group table from Kindergarten Management System source.
MealType	One tuple describes one meal type.	

	ID_MealType	Meal type Id. Surrogate key - generated by database.
	MealType	Meal type. Allowed values: breakfast, lunch, snack, dinner. Taken from MealType from MealRecipe table in Kindergarten Management System source.
MealRecipe	One tuple describes one, specific version of the recipe.	
	ID_MealRecipe	Meal recipe Id. Surrogate key - generated by database. Taken from ID from MealRecipe table in Kindergarten Management System source.
	Name	Meal name. Taken from Name from MealRecipe table in Kindergarten Management System source.
	Version	Version of recipe. Allowed values: ('small', 'big' ...). Established by the recipe having the same Ingredients (<i>Ingredients.name</i>) BUT different Amounts of those Ingredients (<i>IngredientAmount.Amount</i>) in Kindergarten Management System source.
IngredientAmount	One tuple describes one fact of usage of a specific ingredient in a specific recipe.	
	ID_Ingredient	Ingredient Id. Foreign key from Ingredient dimension table
	ID_MealRecipe	Meal recipe Id. Foreign key from MealRecipe dimension table taken from ID from MealRecipe table in Kindergarten Management System source.
Ingredient	One tuple describes one ingredient, of a specific type.	
	ID_Ingredient	Ingredient Id. Surrogate key - generated by database.
	Name	Ingredient name. Taken from Column 3 <i>Name</i> from Order list CSV file source.
	Type	Type of ingredient. Allowed values: dairy, meat, fruit, vegetable, side/other. Taken from Column 2 <i>Type</i> from Order list CSV file source.
MealForChild	One tuple describes one meal for a specific child.	

	ID_Children	Children Id. Foreign key from Children dimension table.
	EatenMealNo	Number of meal eaten. Number generated by the database.
Children	One tuple describes one child.	
	ID_Children	Children Id. Surrogate key - generated by database.
	Pesel	Pesel number of the child. Taken from Pesel from Children table in Kindergarten Management System source.
	FullName	Name and surname of the child. Taken from FullName from Children table in Kindergarten Management System source.
	Age	Age category. Allowed values: 3yr, 4yr, 5yr, 6yr. Determined from DateOfBirth from Children table from Kindergarten Management System source.
	Allergies	Does child have any allergies? Possible answers: 'Yes', 'No'. Determined from Allergies from Children table from Kindergarten Management System source.
	SpecialNeeds	Does child have any special needs? (Like medication). Possible answers: 'Yes', 'No'. Taken from SpecialNeeds from Children table from Kindergarten Management System source.
	ID_GroupName	Group name Id. To which group the child is assigned? Foreign key from Group dimension table. Based on GroupName stored in Group table in Kindergarten Management System source.
MealWithTeacher	One tuple describes one meal with one specific teacher.	
	ID_Teacher	Teacher Id. Foreign key from Teacher dimension table.
	EatenMealNo	Number of meal eaten. Number generated by the database.
Teacher	One tuple describes one teacher.	

	ID_Teacher	Teacher Id. Surrogate key - generated by database.
	Pesel	Pesel number of the teacher. Taken from Pesel from Teacher table in Kindergarten Management System source.
	FullName	Name and surname of the teacher. Taken from FullName from Teacher table in Kindergarten Management System source.
	Qualification	Qualification of the teacher, education status. Taken from Qualification from Teacher table in Kindergarten Management System source.
	Date_first_qualification	Date of first qualification.
	Date_new_qualification	If qualification changes, fill in this date.
	ID_GroupName	Group name Id. Which group is the teacher supervising? Foreign key from Group dimension table. Based on GroupName stored in Group table in Kindergarten Management System source.